LOGARITHMS

Properties of the logarithmic function

Most of them hold for every base; two depend on whether the base is above or below 1.

✓ EVERY BASE ◐ DEPENDS ON BASE

Domain — ✓ EVERY BASE

$$x > 0$$

the argument must be strictly positive

Check that the argument is positive before accepting any solution.

→ Domain

Range — ✓ EVERY BASE

$$(-\infty, \infty)$$

every real value is reached

Monotonicity — ◐ DEPENDS ON BASE

$$\text{increasing for } a > 1$$

decreasing for $0 < a < 1$

→ Monotonicity

The direction of an inequality between logarithms is set by the base.

One-to-one — ✓ EVERY BASE

$$\log_a M = \log_a N \Rightarrow M = N$$

Drop matching logarithms to turn a logarithmic equation into an algebraic one.

→ One-to-one property

Continuity — ✓ EVERY BASE

$$\text{continuous on } (0, \infty)$$

no jumps, breaks or holes

Vertical asymptote — ◐ DEPENDS ON BASE

$$x = 0$$

the sign of the infinity depends on the base

near 0 it drops toward $-\infty$ when $a > 1$ and climbs toward $+\infty$ when the base is below 1; the graph never touches the y-axis

→ Asymptotic behavior

Inverse of the exponential — ✓ EVERY BASE

$$\log_a(a^x) = x,\ a^{\log_a x} = x$$

switch between exponential and logarithmic form

→ Inverse relationship